Marwan Mahmoud

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EXPERIENCE

Software Member - Vision & Autonomy · *2025 – Present*

Triton ROVs · MATE ROV Competition · Regional Informatics Center at AASTMT · Regional Champions & 3rd Place International (Canada)

Perception & State Estimation

- Built full VIO pipeline with ESKF, IMU compensation, TSDF; 10 FPS real-time on edge hardware underwater **Source code**
- Deployed YOLOv8 on MyriadX VPU via OpenVINO; resolved CMX memory overflow and anchor mismatch; identified and documented YOLOv11 architectural incompatibility with MyriadX attention blocks
- Built 3D measurement tool with stereo depth + intrinsic matrices; real-time ±cm uncertainty in high-noise scenes Source code

Streaming & Systems

- Lock-free multi-camera streaming across 5 simultaneous feeds on Raspberry Pi 4, keeping CPU below 70% via Broadcom ISP offloading, thread-safe deque state, and core affinity pinning (os.sched_setaffinity)
- Designed dual-protocol stack (zero-copy binary video, asyncio WebSockets, REST polling); eliminated blocking under concurrent load
- Architected the full software system that performed reliably under competition conditions, directly contributing to the Regional Championship win and Canada qualification
  Source code

Pilot HUD

- Built **low-latency tactical HUD** (HTML5 Canvas, JS, async polling) with freeze-frame, magnification, and crosshairs; designed and tested for **high-latency underwater conditions** (~500ms round-trip)

Python · C++ · asyncio · DepthAI · OpenCV · Open3D · OpenVINO · EKF/ESKF · Linux · WebSockets

Data Analysis Intern · *Summer 2025*

Amreya Petroleum Refining Company (APRC)

- Extracted 50K+ sensor records from SCADA logs; identified 8 anomalies; saved 4 hours/week manual monitoring
- Built statistical filters; generated weekly reports; supported 2 optimization decisions; improved monitoring by 15%

Python · SQL · Data visualization tools

PROJECTS

- **Custom STL Library (C++):** Built 8+ STL containers (vector, linked list, stack, hash map, and sorting algorithms) with templates and iterators; 50+ unit tests. Source code

- **Neural Network from scratch (C++):** designed a PyTorch-style Tensor class with generic autograd engine on top of the custom STL data structure library, then trained the resulting framework on MINST for handwritten digit recognition. Source code | Demo

- **University Registration System (Java, SQL):** Built **concurrent course registration system** handling **10+ simultaneous users**; implemented **MVC design pattern**, **relational schema** (normalized to 3NF), and **thread-safe transaction handling** to prevent race conditions and data corruption. Source code

- **Algorithmic Game AI (Python):** Built a decision-optimal Tic-Tac-Toe agent using game-theoretic search to evaluate outcome pathways. Source code

- **NLP Mathematical Expression Parser (Python):** Built recursive descent parser with tokenization and semantic analysis to interpret plain English math expressions (e.g., "three times x plus five"); supports variable binding and nested expressions, tested on 30+ input cases. Source code

- **Real-Time Sorting Visualizer (C):** Built interactive **algorithm visualization tool** comparing 6 sorting algorithms (**quicksort, mergesort, heapsort, bubblesort, insertion sort, selection sort**) with **live performance metrics** (comparisons, swaps, runtime); demonstrates **time complexity differences** visually. Source code

SKILLS

- **Languages**: Python · C++ · C · Java
- **ML/AI & Data:** NumPy, CuPy, Pandas, TensorFlow, PyTorch, OpenCV, statistical filtering, data pipelines, image processing
- **Embedded & Systems**: embedded C · Linux · asyncio · DepthAI · OpenVINO · EKF/ESKF
- **Tools & Platforms:** Git/GitHub · SQL/SQLite · Docker · WebSockets
- **Core Strengths:** Analytical problem-solving under pressure, quantitative reasoning, structured documentation

EDUCATION

Arab Academy for Science, Technology & Maritime Transport (AASTMT) *B.Sc · Expected May 2028 · GPA 3.45/4.0*

Honors*:* 1st Place, Regional Champion - MATE ROV Egypt Competition (April 2026) · 3rd Place - MATE ROV International Competition, Canada (2026) · Overall Score: 716.33 and Best Engineering Presentation, Best Technical Documentation Awards.

Relevant Coursework*: **Data Structures & Algorithms** · OOP · DSP · Electrical Circuits · Electronics · Digital Logic Design · Calculus · Deferential Equations · Vector Algebra · Linear Algebra*

COURSES & CERTIFICATES

- CS50x, CS50Python, & CS50SQL from Harvard University.
- Supervised Learning, Regression and Classification by Andrew Ng.
- Advanced Learning Algorithms by Andrew Ng (Ongoing).

LANGUAGES

Arabic (Native) · English (Fluent, C1)